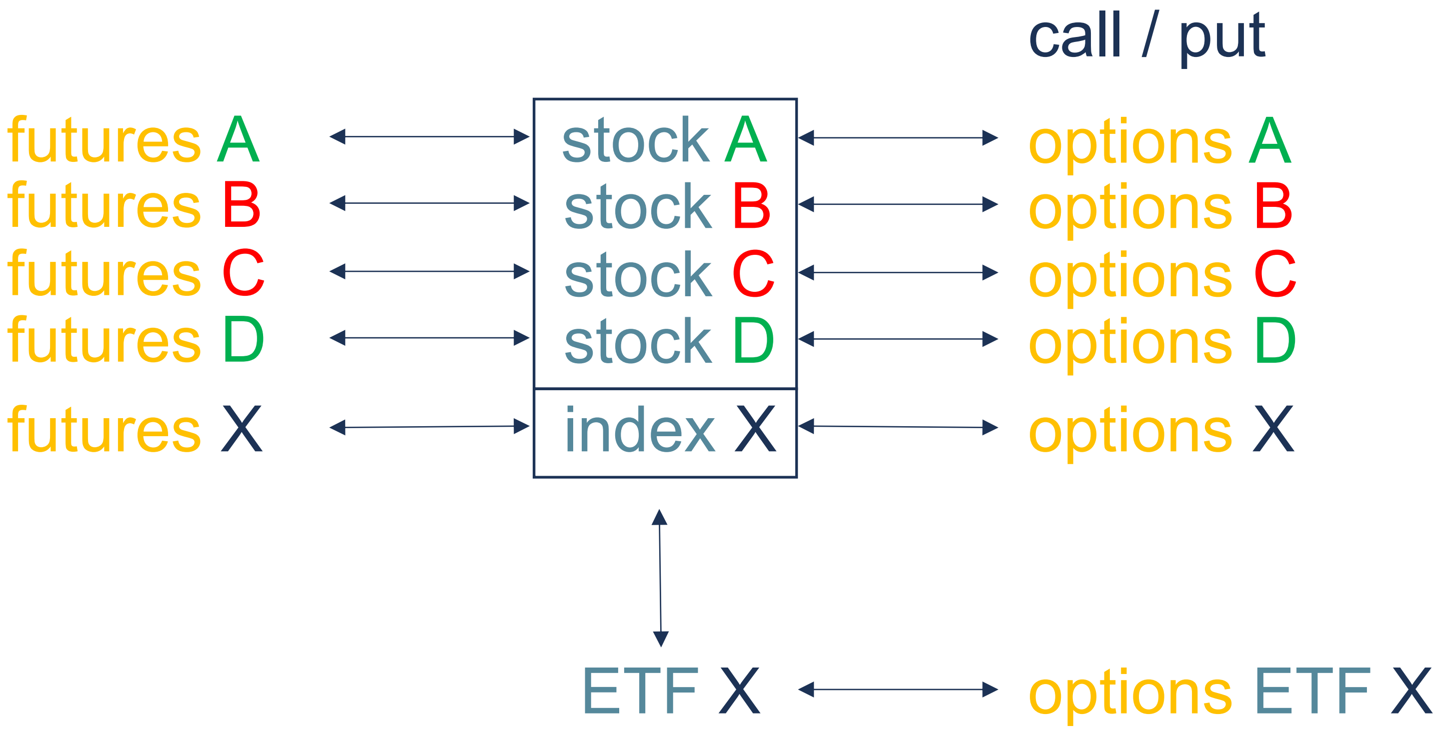
Optiver

Market Making strategies

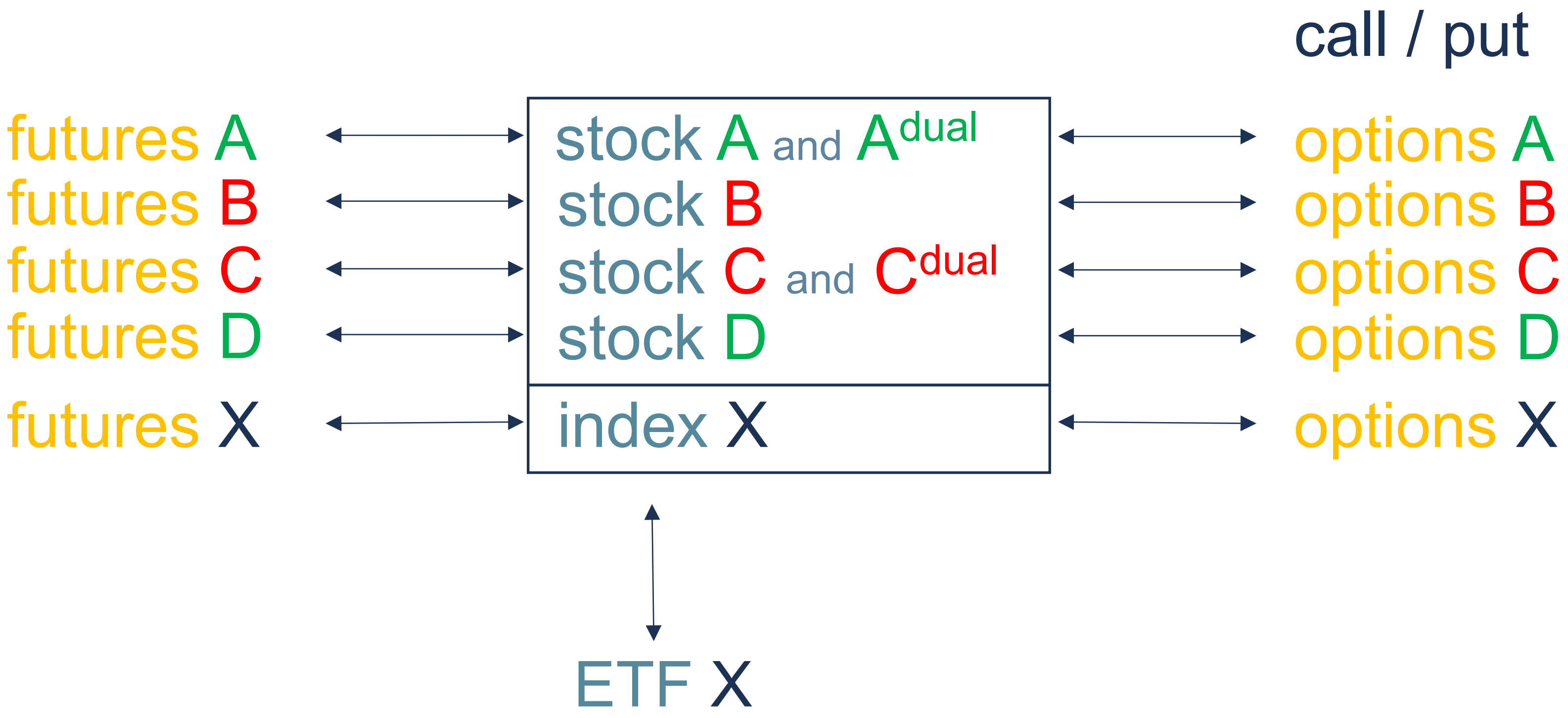
Market making trading strategies

- Dual / multiple stock listing
- Stock – future
- ETF – future
- Stock – options
- Index – options
- Correlations

Index complex



Dual / multiple stock listing



Dual / multiple stock listing

stock A and A^{dual}

stock C and C^{dual}

Dual listing

Stock A

bid volume	price	ask volume
	44.67	8539
	44.66	10551
	44.65	3677
	44.64	6800
	44.63	9061
	44.62	6693
	44.61	6573
2049	**44.60**	
5176	**44.59**	
1080	**44.58**	
5000	**44.57**	
16776	**44.56**	
2000	**44.55**	
15349	**44.54**	
7800	**44.53**	
11327	**44.52**	

Stocks exchange X

1

price of Stock A is input for market in dual Stock A

dual $S_A = S_A$

after a trade in dual Stock A, a hedge order in Stock A needs to be executed

2

dual Stock A

bid volume	price	ask volume
	44.67	560
	44.66	
	44.65	230
	44.64	178
	44.63	
	44.62	
	44.61	
	44.60	
	44.59	
	44.58	
	44.57	
236	**44.56**	
	44.55	
450	**44.54**	
	44.53	
641	**44.52**	

Stocks exchange Y

Index complex

				call / put
futures A	↔	stock A	↔	options A
futures B	↔	stock B	↔	options B
futures C	↔	stock C	↔	options C
futures D	↔	stock D	↔	options D
futures X	↔	index X	↔	options X
		↕		
		ETF X	↔	options ETF X

Stock - future

futures A ⟷ stock A

Stock - future

Stock A

bid volume	price	ask volume
	44.67	8539
	44.66	10551
	44.65	3677
	44.64	6800
	44.63	9061
	44.62	6693
	44.61	6573
2049	**44.60**	
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2000	**44.55**	
15349	**44.54**	
7800	**44.53**	
11327	**44.52**	

Stocks exchange

1

price of Stock A is input for market in Future A

F = f(S,i,d,τ)

after a trade in Future A, a hedge order in Stock A needs to be executed

2

Future A

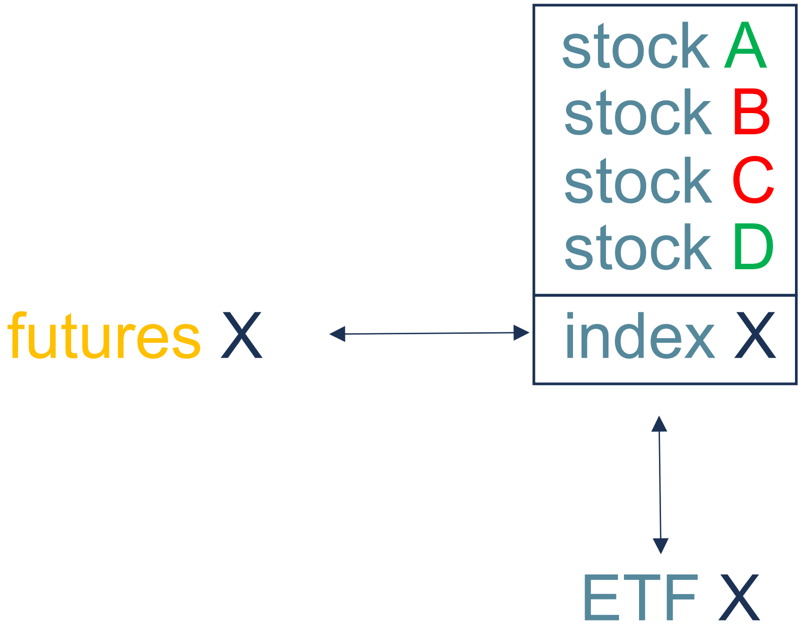
bid volume	price	ask volume
	45.36	346
	45.35	290
	45.34	240
	45.33	139
	45.32	65
	45.31	16
	45.30	2
	45.29	
15	**45.28**	
62	**45.27**	
135	**45.26**	
289	**45.25**	
361	**45.24**	
490	**45.23**	
528	**45.22**	
570	**45.21**	

Futures exchange

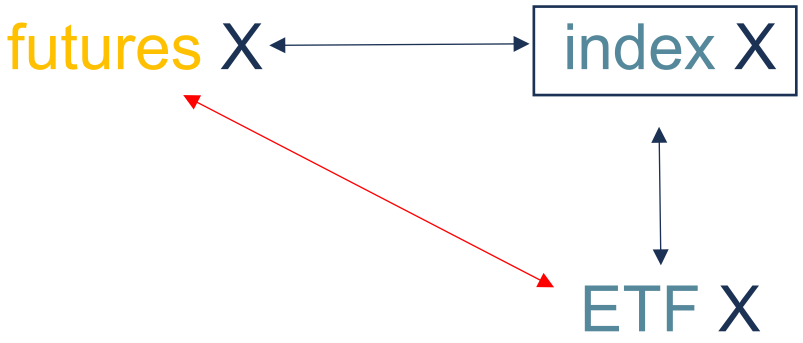
Index complex

				call / put
futures A	↔	stock A	↔	options A
futures B	↔	stock B	↔	options B
futures C	↔	stock C	↔	options C
futures D	↔	stock D	↔	options D
futures X	↔	index X	↔	options X
		↕		
		ETF X	↔	options ETF X

Index future - ETF



Index future - ETF



Future - ETF

index Future

bid volume	price	ask volume
	372.40	2600
	372.35	2180
	372.30	1642
	372.25	1475
	372.20	1263
	372.15	908
	372.10	789
590	**372.05**	
620	**372.00**	
783	**371.95**	
1029	**371.90**	
1268	**371.85**	
1690	**371.80**	
1800	**371.75**	
2377	**371.70**	
3206	**371.65**	

Futures exchange

1

price of index Future is input for market in ETF

ETF = f(F,i,d,τ,cc)

after a trade in ETF, a hedge order in index Future needs to be executed

2

ETF

bid volume	price	ask volume
	37.52	3745
	37.51	2300
	37.50	1450
	37.49	838
	37.48	731
	37.47	460
	37.46	342
200	**37.45**	
610	**37.44**	
1200	**37.43**	
1820	**37.42**	
2682	**37.41**	
3204	**37.40**	
3600	**37.39**	
4800	**37.38**	
7305	**37.37**	

Stocks exchange

Index complex

call / put

futures A ⟷ stock A ⟷ options A
futures B ⟷ stock B ⟷ options B
futures C ⟷ stock C ⟷ options C
futures D ⟷ stock D ⟷ options D
futures X ⟷ index X ⟷ options X

ETF X ⟷ options ETF X

Stock - options

call / put

stock A ⟷ options A

Stock - options

Stock A

bid volume	price	ask volume
	44.67	8539
	44.66	10551
	44.65	3677
	44.64	6800
	44.63	9061
	44.62	6693
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15349	**44.54**	
7800	**44.53**	
11327	**44.52**	

Stocks exchange

1

price of Stock A is input for market in Option A

Option = f(S,X,i,τ,d,σ)

after a trade in Option A, a hedge order in Stock A needs to be executed

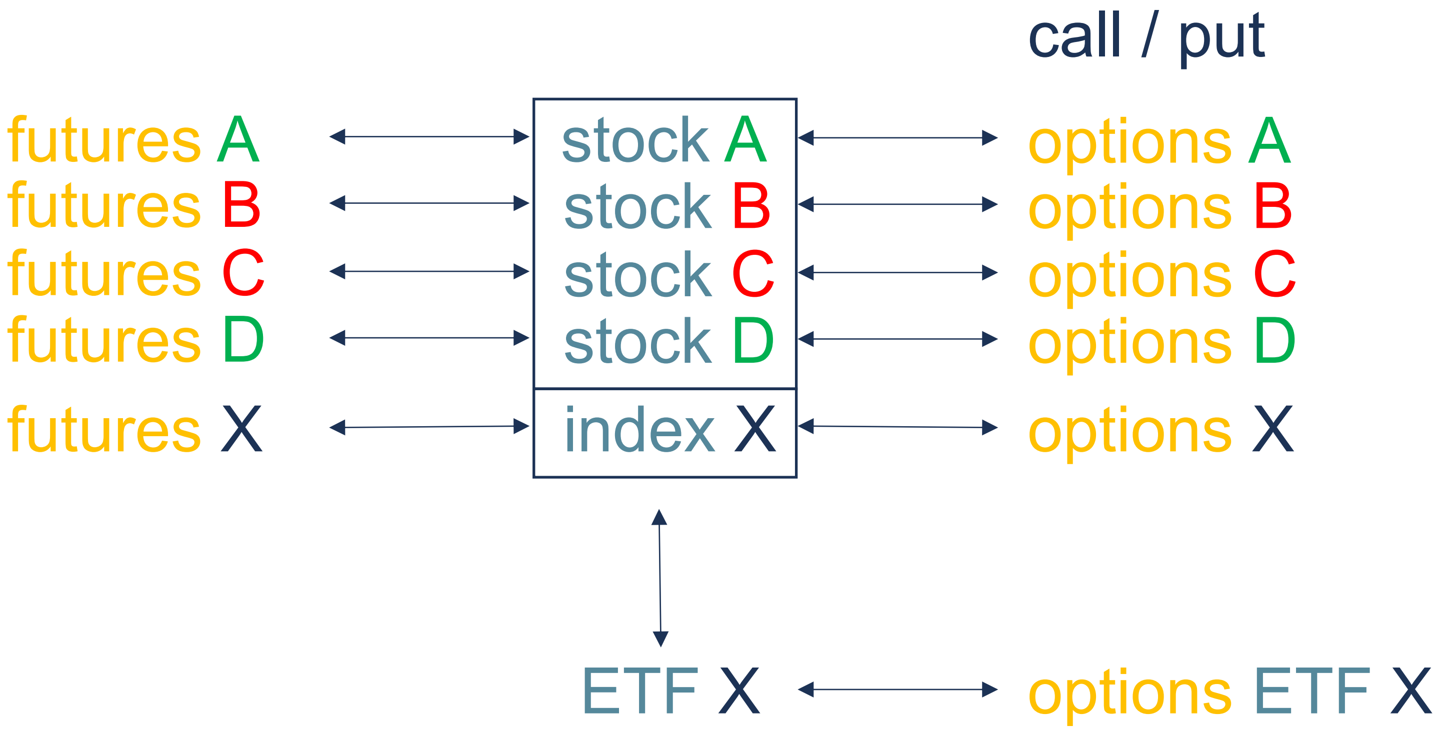
2

Option A

bid volume	price	ask volume
	11.60	200
	11.55	160
	11.50	
	11.45	23
	11.40	27
	11.35	
	11.30	
	11.25	
	11.20	
	11.15	
25	**11.10**	
40	**11.05**	
	11.00	
38	**10.95**	
	10.90	
148	**10.85**	

Options exchange

Index complex



Index complex

call / put

stock A
stock B
stock C
stock D

futures X ⟷ index X ⟷ options X

Index complex

call / put

futures X ⟷ options X

Index - options

Index future

bid volume	price	ask volume
	372.40	2600
	372.35	2180
	372.30	1642
	372.25	1475
	372.20	1263
	372.15	908
	372.10	789
590	372.05	
620	372.00	
783	371.95	
1029	371.90	
1268	371.85	
1690	371.80	
1800	371.75	
2377	371.70	
3206	371.65	

Futures exchange

1

price of index Future is input for market in index option

Option = f(F,X,i,τ,d,σ)

after a trade in the index Option, a hedge order in index Future needs to be executed

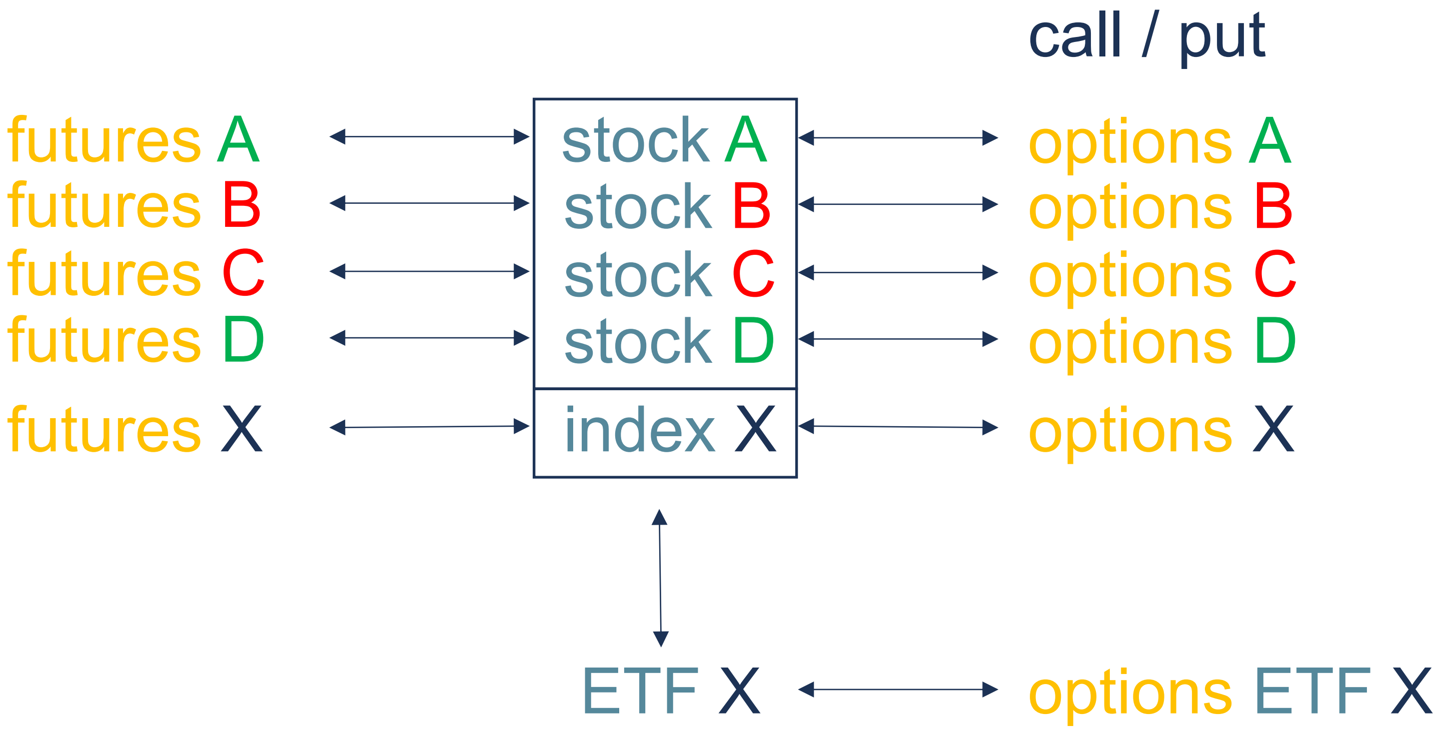
2

Index option

bid volume	price	ask volume
	64.50	312
	64.40	250
	64.30	242
	64.20	190
	64.10	120
	64.00	46
	63.90	32
	63.80	
20	63.70	
130	63.60	
150	63.50	
189	63.40	
235	63.30	
250	63.20	
350	63.10	
521	63.00	

Options exchange

Index complex



Correlation

stock A

stock C

Correlation

Instrument A

bid volume	price	ask volume
	44.67	8539
	44.66	10551
	44.65	3677
	44.64	6800
	44.63	9061
	44.62	6693
	44.61	6573
2049	**44.60**	
5176	**44.59**	
1080	**44.58**	
5000	**44.57**	
16776	**44.56**	
2000	**44.55**	
15349	**44.54**	
7800	**44.53**	
11327	**44.52**	

Exchange X

1

price of instrument A is input for market in instrument C

instr C = f(instr A)

after a trade in instrument C, a hedge order in instrument A needs to be executed

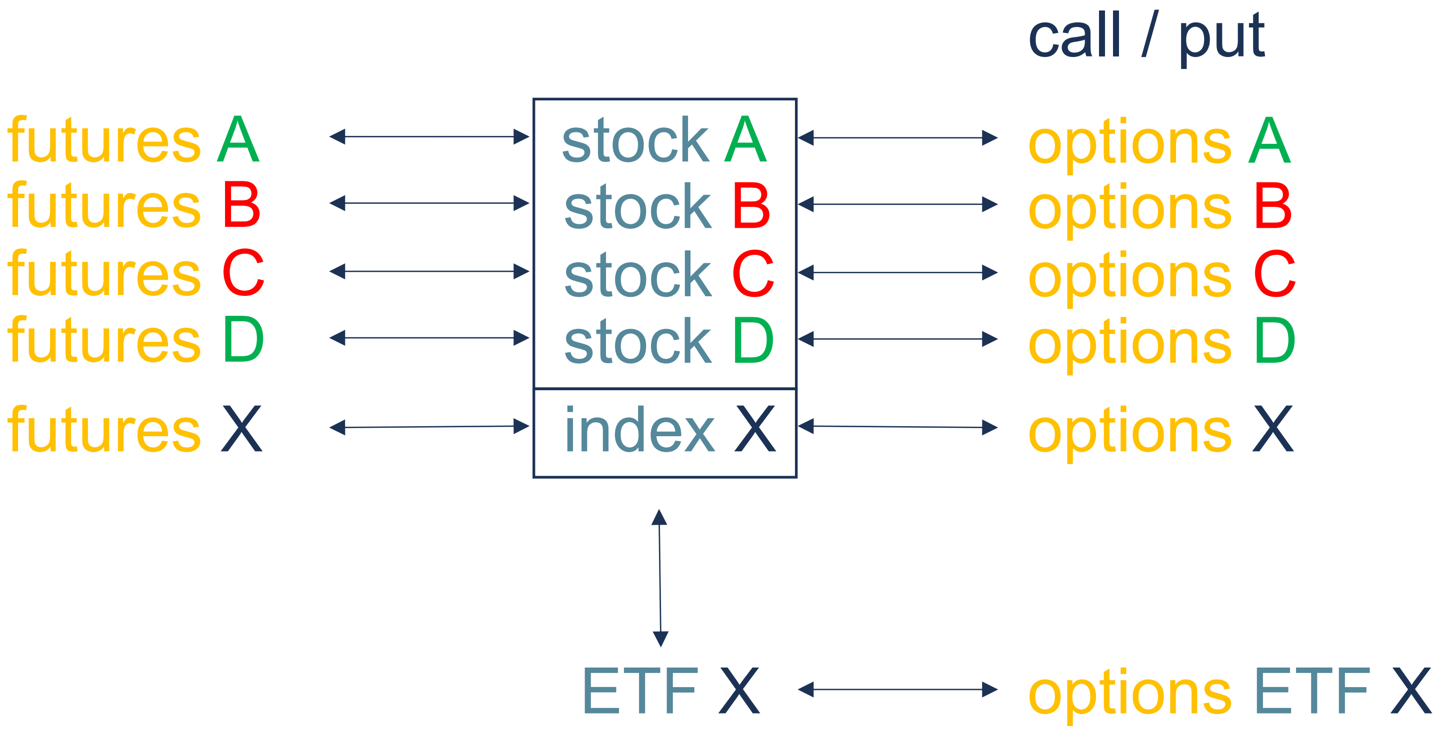
2

Instrument C

bid volume	price	ask volume
	37.51	917
	37.50	742
	37.49	529
	37.48	534
	37.47	422
	37.46	282
	37.45	112
279	**37.44**	
346	**37.43**	
501	**37.42**	
521	**37.41**	
563	**37.40**	
599	**37.39**	
672	**37.38**	
892	**37.37**	
1274	**37.36**	

Exchange Y

Index complex



Optiver

Thank you!

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